

Architectural standard for software systems

Estándar Arquitectónico para Sistemas de Software

First Edition

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Main Author / Autora Principal:

Dr. Paula Melissa Alfaro Montero, *Clinical Microbiologist & AI Governance Architect*

ORCID: 0009-0004-2849-5794

Entity / Entidad: LUNA Research / Global Sigma Initiative

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⚡ QUICK REFERENCE / REFERENCIA RÁPIDA

Quick Reference Card

A one-page operational summary of the Standard. / Resumen operacional de una página del Estándar.

5 Non-Negotiable Principles / 5 Principios No Negociables

P1

TRACEABILITY / TRAZABILIDAD

Every system behavior must be explainable by observable artifacts (events, logs, state). No action without a trace.

P2

STATE CONSISTENCY / CONSISTENCIA DE ESTADO

The system must maintain a consistent representation of its state. State must be reconstructible or explicitly stored.

P3

VERIFIABILITY / VERIFICABILIDAD

System behavior must be testable against defined invariants. Every invariant must have at least one validation mechanism.

P4

GOVERNANCE / GOBERNANZA

Decision authority must be explicitly defined: which agents can make which decisions, and when human intervention is required.

P5

BOUNDED CAPABILITY / CAPACIDAD ACOTADA

The system must explicitly define what it cannot or should not do. These boundaries are part of its identity and are auditable.

6 Required Documents / 6 Documentos Requeridos

#	Document / Documento	Purpose / Propósito	Mandatory / Obligatorio
1	Canonical Design	System identity, entities, invariants, rules, boundaries / Identidad del sistema	✓ YES

#	Document / Documento	Purpose / Propósito	Mandatory / Obligatorio
2	Architecture Brief (Technical)	Operational flow, roles, interactions, state transitions / Flujo operacional	✓ YES
3	SSOT	Technology stack, exact versions, environments / Stack tecnológico exacto	✓ YES
4	C4 Model	Architectural visualization L1-L3 / Visualización arquitectónica	✓ YES
5	SDD	Interfaces, schemas, APIs, invariant tests / Interfaces y pruebas	✓ YES
6	VCL	Invariant tests, audit mechanisms, compliance level / Evidencia de auditoría	✓ YES

Compliance Levels / Niveles de Conformidad

Level	Name / Nombre	Requirements / Requisitos
CL-0	Conceptual	All 6 documents exist at conceptual level / Los 6 documentos existen al nivel conceptual
CL-1	Fully Documented	All documents with sufficient detail / Todos los documentos con detalle suficiente
CL-2	Verifiable	CL-1 + automated invariant tests + full traceability / CL-1 + pruebas automatizadas + trazabilidad completa
CL-3	Governed	CL-2 + HITL + escalation + third-party reproducible audit / CL-2 + gobernanza activa + auditoría reproducible

⚠ Absence of any required document = Automatic CL-0 or audit rejection. / La ausencia de cualquier documento = CL-0 automático o rechazo de auditoría.

PART 1 — NORMATIVE BASE / PARTE 1 — BASE NORMATIVA

0. Purpose / Propósito

EN: This document defines a technical standard for the design, validation, and audit of software system architectures, independent of specific implementations.

ES: Este documento define un estándar técnico para el diseño, validación y auditoría de arquitecturas de sistemas de software, independiente de implementaciones específicas.

This standard:

- Enables coherent architectural design across systems.
- Provides a normalized structure for documentation.
- Enables pre-implementation auditability.
- Supports governance, certification, and scalability.

Este estándar:

- Habilita diseño arquitectónico coherente entre sistemas.
- Provee estructura normalizada para documentación.
- Habilita auditabilidad pre-implementación.
- Soporta gobernanza, certificación y escalabilidad.

Although originally conceived with emphasis on cognitive systems, this standard is fully applicable to traditional software systems. / Aunque concebido con énfasis en sistemas cognitivos, este estándar es plenamente aplicable a sistemas de software tradicionales.

Relation to the Canonical Design Guide / Relación con la Guía de Diseño Canónico

The Canonical Design Guide is a reference implementation of this standard for cognitive systems. It demonstrates full compliance with all requirements defined herein, but it is not the only valid implementation. Any system that satisfies the requirements of this standard is conformant.

La Guía de Diseño Canónico es una implementación de referencia de este estándar para sistemas cognitivos. Demuestra conformidad total con todos los requisitos, pero no es la única implementación válida. Cualquier sistema que satisfaga los requisitos de este estándar es conforme.

1. Scope / Alcance

This standard applies to any software system that requires explicit, governable, and verifiable design. It includes, but is not limited to:

Este estándar aplica a cualquier sistema de software que requiera diseño explícito, gobernado y verificable. Incluye, entre otros:

System Type / Tipo de Sistema	Examples / Ejemplos
Cognitive systems / Sistemas cognitivos	AI/LLMs, autonomous reasoning, RAG, knowledge graphs
Traditional systems / Sistemas tradicionales	CRUD, embedded, batch, services, middleware
Hybrid systems / Sistemas híbridos	Combinations of the above / Combinaciones de los anteriores
Any system requiring traceability & governance	Financial, medical, legal, critical infrastructure

Technological assumptions / Supuestos tecnológicos:

- Traceability is possible (via events, logs, state, or equivalent).
- State can be represented consistently.
- System behavior can be evaluated against defined criteria.

This standard is technologically agnostic. It prescribes no technologies, languages, or concrete implementations. / Este estándar es agnóstico tecnológicamente. No prescribe tecnologías, lenguajes ni implementaciones concretas.

1.1 Applicability and Adoption Scenarios / Escenarios de Aplicación y Adopción

This standard is designed for systems where governability, verifiability, and traceability are required or desired. It is not mandatory for every software system. The following scenarios illustrate how the standard can be applied according to the system's nature and needs.

Este estándar está diseñado para sistemas donde se requiere o desea gobernabilidad, verificabilidad y trazabilidad. No es obligatorio para todo sistema de software. Los siguientes escenarios ilustran cómo se puede aplicar el estándar según la naturaleza y necesidades del sistema.

Scenario / Escenario	Applicability / Aplicabilidad	Recommended Approach / Enfoque Recomendado
Prototype or throwaway code / Prototipo o código desechable	Out of scope / Fuera de alcance	Not required. May use minimal principles (e.g., purpose statement) if useful. / No requerido. Puede usar principios mínimos si es útil.
Internal tool with low impact / Herramienta interna de bajo impacto	Optional / Opcional	Partial adoption: basic documentation (CL-0 or CL-1) to capture design intent. / Adopción parcial: documentación básica (CL-0 o CL-1) para capturar la intención de diseño.
System expected to evolve into critical domain / Sistema que evolucionará hacia dominio crítico	Recommended / Recomendado	Start with Canonical Design and CL-1; progressively add governance as criticality increases. / Comenzar con Diseño Canónico y CL-1; agregar gobernanza progresivamente.
System with regulatory or high-risk requirements / Sistema con requisitos regulatorios o de alto riesgo	Mandatory / Obligatorio	Full adoption targeting CL-2 or CL-3 from the start. / Adopción completa apuntando a CL-2 o CL-3 desde el inicio.
Mixed system (critical and non-critical modules) / Sistema mixto (módulos críticos y no críticos)	Modular / Modular	Apply the standard only to critical modules; others may follow lighter practices. / Aplicar el estándar solo a los módulos críticos; otros pueden seguir prácticas más ligeras.

The standard does not impose a one-size-fits-all mandate. Organizations are free to adopt it in whole or in part, and to scale their level of compliance according to the actual risk and governance needs of each system.

El estándar no impone un mandato único para todos. Las organizaciones son libres de adoptarlo total o parcialmente, y de escalar su nivel de conformidad según el riesgo real y las necesidades de gobernanza de cada sistema.

2. Fundamental Principles (Non-Negotiable) / Principios Fundamentales (No Negociables)

Every system conformant with this standard must satisfy the following five principles. These are binding regardless of technology, domain, or implementation approach.

Todo sistema conforme con este estándar debe satisfacer los siguientes cinco principios. Son vinculantes independientemente de tecnología, dominio o enfoque de implementación.

P1

TRACEABILITY / TRAZABILIDAD

Every system behavior must be explainable by observable artifacts (events, logs, state, or equivalent). No action without a trace. / Todo comportamiento del sistema debe ser explicable por artefactos observables. Ninguna acción sin traza.

P2

STATE CONSISTENCY / CONSISTENCIA DE ESTADO

The system must maintain a consistent representation of its state. State must be reconstructible or explicitly stored, and changes must be traceable. / El sistema debe mantener representación consistente de su estado. El estado debe ser reconstruible o explícitamente almacenado.

P3

VERIFIABILITY / VERIFICABILIDAD

System behavior must be testable against defined invariants. Every invariant must have at least one validation mechanism. / El comportamiento del sistema debe ser testeable contra invariantes definidos. Cada invariante debe tener al menos un mecanismo de validación.

P4

GOVERNANCE / GOBERNANZA

Decision authority must be explicitly defined: which agents can make which decisions, and when human intervention is required. / La autoridad de decisión debe ser explícitamente definida: qué agentes pueden tomar qué decisiones, y cuándo se requiere intervención humana.

P5

BOUNDED CAPABILITY / CAPACIDAD ACOTADA

The system must explicitly define what it cannot or should not do. These boundaries are part of its identity and are auditable. / El sistema debe definir explícitamente lo que no puede o no debe hacer. Estos límites son parte de su identidad y son auditables.

3. Required Architectural Documents / Documentos Arquitectónicos Requeridos

Every system aspiring to conform to this standard MUST provide the following documents. The absence of any of them implies automatic non-conformity or classification at the lowest compliance level (CL-0).

Todo sistema que aspire a conformar con este estándar DEBE proveer los siguientes documentos. La ausencia de cualquiera implica no conformidad automática o clasificación en el nivel mínimo (CL-0).

3.1 Canonical Design Document / Documento de Diseño Canónico

Defines the essence of the system — the single source of truth for the system's logic.

Define la esencia del sistema — la única fuente de verdad para la lógica del sistema.

Minimum content / Contenido mínimo:

- System purpose / Propósito del sistema
- Fundamental entities / Entidades fundamentales
- Functional and verification invariants / Invariantes funcionales y de verificación
- Behavior rules / Reglas de comportamiento
- Capability boundaries / Límites de capacidad

3.2 Architecture Brief (Technical) / Brief Arquitectónico (Técnico)

Instantiates the canonical design in a concrete functional context.

Instancia el diseño canónico en un contexto funcional concreto.

Minimum content / Contenido mínimo:

- Operational flow / Flujo operacional
- Roles of agents or modules / Roles de agentes o módulos
- Interaction patterns / Patrones de interacción
- State transitions / Transiciones de estado
- Functional configuration (thresholds, modes, policies) / Configuración funcional
- Boundary model / Modelo de límites

3.3 SSOT (Single Source of Truth) / SSOT (Fuente Única de Verdad)

Concrete technological decisions.

Decisiones tecnológicas concretas.

Minimum content / Contenido mínimo:

- Technology stack per layer (languages, frameworks, databases, buses) / Stack tecnológico por capa
- Exact versions / Versiones exactas
- Environments (development, testing, production) / Entornos
- Configuration constraints / Restricciones de configuración
- Justification for each choice / Justificación de cada decisión

3.4 C4 Model / Modelo C4

Architectural visualization.

Visualización arquitectónica.

Minimum content / Contenido mínimo:

- Context Diagram (Level 1) / Diagrama de Contexto (Nivel 1)
- Container Diagram (Level 2) with technology names / Diagrama de Contenedores con nombres tecnológicos
- Component Diagram (Level 3) for critical containers / Diagrama de Componentes para contenedores críticos

3.5 SDD (Software Design Document)

Detailed implementation specification.

Especificación de implementación detallada.

Minimum content / Contenido mínimo:

- Interfaces
- Data schemas (events, objects) / Esquemas de datos
- Event structures / Estructuras de eventos
- APIs
- Configuration logic / Lógica de configuración
- Verification tests with invariant coverage / Pruebas de verificación con cobertura de invariantes

3.6 Verification & Compliance Layer (VCL) / Capa de Verificación y Conformidad

Evidence that the system is correct and auditable.

Evidencia de que el sistema es correcto y auditable.

Minimum content / Contenido mínimo:

- Invariant tests — each invariant must have at least one test / Pruebas de invariantes
- Validation rules / Reglas de validación
- Audit mechanisms / Mecanismos de auditoría
- Compliance level achieved (CL-0 to CL-3) / Nivel de conformidad alcanzado
- Coverage evidence / Evidencia de cobertura

4. Mandatory Invariant Structure / Estructura Mandatoria de Invariantes

Every system must define two types of invariants:

Todo sistema debe definir dos tipos de invariantes:

Type / Tipo	Purpose / Propósito
Functional / Funcional	Define what the system must always do / Definen lo que el sistema debe hacer siempre
Verification / De verificación	Define how correctness is validated / Definen cómo se valida la corrección

Coverage Rule / Regla de Cobertura

Each invariant MUST:

- Have a unique identifier. / Tener un identificador único.
- Be testable (via test, manual validation, or equivalent mechanism). / Ser testeable.
- Have at least one documented validation mechanism. / Tener al menos un mecanismo de validación documentado.

⚠ The absence of a test for an invariant implies automatic non-conformity with CL-2 or higher levels. / La ausencia de prueba para un invariante implica no conformidad automática con CL-2 o superior.

5. State Model Requirements / Requisitos del Modelo de Estado

Systems must explicitly define how state is created, modified, managed, and kept consistent.

Los sistemas deben definir explícitamente cómo se crea, modifica, gestiona y mantiene consistente el estado.

Requirement / Requisito	Description / Descripción
Creation / Creación	How state is created / Cómo se crea el estado
Modification / Modificación	How state is modified / Cómo se modifica el estado
Authority / Autoridad	Who has authority over state / Quién tiene autoridad sobre el estado

Requirement / Requisito	Description / Descripción
Consistency / Consistencia	How consistency is maintained / Cómo se mantiene la consistencia
Reconstructibility / Reconstruibilidad	State reconstructible from events OR explicitly stored / Reconstruible desde eventos O almacenado explícitamente
Traceability / Trazabilidad	Changes traceable: who, when, why / Cambios trazables: quién, cuándo, por qué
Conflict resolution / Resolución de conflictos	Explicit rules for state conflicts / Reglas explícitas para conflictos

Recommended: Append-only model with obsolescence marking over physical deletion. / Recomendado: Modelo de solo-adición con marcado de obsolescencia en lugar de eliminación física.

6. Response Model Requirements / Requisitos del Modelo de Respuesta

System outputs must be derivable from system state or inputs, must not contradict known state, and must declare uncertainty when applicable.

Las salidas del sistema deben ser derivables del estado o entradas, no deben contradecir el estado conocido, y deben declarar incertidumbre cuando aplica.

Principle of Response Derivation / Principio de Derivación de Respuesta

Every response must be a determinable function of the reconstructible state, within declared uncertainty bounds. No generation of responses that cannot be explained as a transformation of verifiable state is permitted.

Toda respuesta debe ser una función determinable del estado reconstruible, dentro de límites de incertidumbre declarados. No se permite generación de respuestas que no puedan explicarse como transformación de estado verificable.

7. Governance Requirements / Requisitos de Gobernanza

Systems must explicitly define governance structures. / Los sistemas deben definir explícitamente las estructuras de gobernanza.

Component / Componente	Definition Required / Definición Requerida
Authority levels / Niveles de autoridad	Which decisions each agent can make autonomously / Qué decisiones puede tomar cada agente autónomamente
Human intervention / Intervención humana	When supervision is mandatory / Cuando la supervisión es obligatoria
Escalation rules / Reglas de escalación	What to do when a conflict cannot be resolved automatically / Qué hacer cuando un conflicto no puede resolverse automáticamente

8. Capability Boundaries / Límites de Capacidad

Systems must explicitly define what they cannot or should not do. These boundaries are non-negotiable in any conformant implementation.

Los sistemas deben definir explícitamente lo que no pueden o no deben hacer. Estos límites son no negociables en cualquier implementación conforme.

Boundary Type / Tipo de Límite	Description / Descripción
Prohibited actions / Acciones prohibidas	What the system cannot do / Lo que el sistema no puede hacer
Ethical-functional limits / Límites éticos-funcionales	E.g., 'does not generate knowledge without verifiable basis' / No genera conocimiento sin base verificable
Abstention conditions / Condiciones de abstención	When the system must refuse to act / Cuando el sistema debe abstenerse de actuar

9. Audit Requirements / Requisitos de Auditoría

For a system to be auditable, it must provide:

Para que un sistema sea auditable, debe proveer:

- The complete set of documents defined in Section 3. / El conjunto completo de documentos definidos en la Sección 3.
- Traceability artifacts (logs, events, reconstructible state). / Artefactos de trazabilidad.
- Invariant coverage evidence (executed tests, results). / Evidencia de cobertura de invariantes.
- Validation evidence (reports, metrics, simulations). / Evidencia de validación.

10. Compliance Levels / Niveles de Conformidad

Level / Nivel	Name / Nombre	Requirements / Requisitos
CL-0	Conceptually Documented / Documentado Conceptualmente	All documents in Section 3 exist at a conceptual level / Todos los documentos de Sección 3 existen en nivel conceptual
CL-1	Fully Documented / Completamente Documentado	All documents in Section 3 exist with sufficient detail / Todos los documentos con detalle suficiente
CL-2	Verifiable / Verificable	CL-1 + all invariants have automated tests + complete traceability / CL-1 + pruebas automatizadas + trazabilidad completa
CL-3	Governed / Gobernado	CL-2 + active governance (HITL, escalation) + audit reproducible by third parties / CL-2 + gobernanza activa + auditoría reproducible por terceros

Note / Nota: A system can be conformant to the standard at different levels for different modules or contexts. / Un sistema puede ser conforme al estándar en diferentes niveles para diferentes módulos o contextos.

11. Applicability / Aplicabilidad

This standard can be used for:

- Designing new systems from scratch. / Diseñar nuevos sistemas desde cero.
- Evaluating third-party architectures (technical due diligence). / Evaluar arquitecturas de terceros (due diligence técnico).
- Standardizing internal frameworks in organizations. / Estandarizar frameworks internos en organizaciones.
- Preparing systems for external certification. / Preparar sistemas para certificación externa.

12. Key Outcome / Resultado Clave

A system conformant with this standard is:

Property / Propiedad	EN	ES
Understandable / Comprensible	Its design is explicit and documented	Su diseño es explícito y documentado
Verifiable / Verificable	Its correctness can be checked against invariants	Su corrección puede verificarse contra invariantes
Governable / Gobernado	Decisions are bounded and supervised	Las decisiones están acotadas y supervisadas
Auditable / Auditable	Evidence of all the above exists	Existe evidencia de todo lo anterior

Regardless of whether it is cognitive or traditional. / Independientemente de si es cognitivo o tradicional.

13. Relation to the Canonical Design Guide / Relación con la Guía de Diseño Canónico

This standard defines the minimum requirements that every software system architecture must satisfy.

The Canonical Design Guide is a reference implementation for cognitive systems that demonstrates full compliance with this standard. It provides:

- A concrete model of invariants (INV-1 to INV-12, INV-V1 to INV-V6).
- An event-based agent architecture.
- An explicit state model with defined authority.
- A verification methodology with compliance levels.

Any system that satisfies all requirements of this standard is conformant, regardless of its internal architecture, technologies, or design methodology. / Cualquier sistema que satisfaga todos los requisitos de este estándar es conforme, independientemente de su arquitectura interna, tecnologías o metodología de diseño.

14. Standard Addenda / Addenda del Estándar

Addendum	Title	Purpose / Propósito	Applicability / Aplicabilidad
A	Flow-Based Agent Model	Internal structure of flow-based agents	Systems with agents

Addendum	Title	Purpose / Propósito	Applicability / Aplicabilidad
B	RAE: Architectural Reconstruction	Infer design from artifacts	Legacy or undocumented systems
C	Execution with Agents and Agile Flows	Lifecycle from canonical design to production	Agile development with AI
D	Tooling Ecosystem (Reference)	Categories of tools to implement defined flows	Implementation
E	Metrics and Observability	Metrics to evaluate process and system health	Continuous improvement, audit
F	Knowledge Architecture	RAG, chunking, grounding, knowledge management	Systems with knowledge management
G	Evaluation & Benchmarking	Evaluate, compare, and certify systems	Certification, comparison
H	Runtime Observability	Observability, continuous validation, adaptive governance	Production systems
I	Document Glossary	Common vocabulary for ecosystem documents	Any system
J	Guide to Building Technical Glossaries	Methodology to create system-specific glossaries	Any system
K	Adoption & Certification Frameworks	Progressive adoption, implementation, certification	Any system
L	Universal Adoption & Derivation Framework	Derive all technical and audit documents from Canonical Design	Any system
M	Minimum Document Templates	Skeleton templates for each required document	Any system

Addenda are optional complementary documents. A system can be conformant to the standard without implementing any of them.

15. Request for Information for Audit (RFI/RFP)

Any system submitted for evaluation MUST include:

- Canonical Design Document
- Architecture Brief (Technical)
- SSOT (Single Source of Truth)
- C4 Model (context, containers, components)
- SDD (Software Design Document)
- Verification evidence (tests, coverage, results)

⚠ The lack of any of these documents results in: Automatic rejection of the audit OR Classification at CL-0 with the corresponding validity limitation. / La falta de cualquiera de estos documentos resulta en: Rechazo automático de la auditoría O Clasificación en CL-0.

16. Versioning Policy / Política de Versionado

Type / Tipo	Version / Versión	Example / Ejemplo	Compatibility / Compatibilidad
Major	vX.0	Changes to fundamental principles or structural requirements	Breaking: previous implementations may be non-conformant
Minor	vX.Y	New optional requirements, normative clarifications	Compatible: previous implementations remain valid
Patch	vX.Y.Z	Typographical corrections, examples, references	Fully compatible

This document is in FROZEN state (v1.2). Only patches for errors or clarifications that do not alter structural requirements will be accepted.

17. Change Log / Registro de Cambios

Version	Date / Fecha	Changes / Cambios	Type / Tipo
v1.0	2026-03-18	Initial version / Versión inicial	—
v1.1	2026-03-19	Title changed. Scope extended to traditional systems. Purpose clarified. Section 13 clarified. Addenda A–J added.	Minor
v1.2	2026-03-25	Added Addenda K, L, M. Functional equivalence tables, derivation hierarchy, and minimum document templates. Added Appendix 2 (Global Sigma Alignment). Minor clarifications in Sections 0 and 13.	Minor

ADDENDUM K — Adoption & Certification Frameworks

Addendum K – Adoption & Certification Frameworks

Purpose: Establish structured frameworks for progressive adoption, implementation, and certification of the Standard, ensuring accessibility, scalability, and consistency without compromising normative rigor.

Propósito: Establecer frameworks estructurados para la adopción progresiva, implementación y certificación del Estándar.

Status / Estado: Proposed — In development / Propuesto — En desarrollo

Nature / Naturaleza: Strategic-operational (non-normative, does not alter requirements) / Estratégico-operacional (no normativo, no altera requisitos)

K.1 Guiding Principle / Principio Rector

The depth of the standard is preserved. Adoption is structured. / La profundidad del estándar se preserva. La adopción es estructurada.

K.2 Adoption Framework / Framework de Adopción

K.2.1 Adoption Levels / Niveles de Adopción

Level / Nivel	Name / Nombre	Description / Descripción	Outcome / Resultado
Level 0	Exploration / Exploración	Conceptual understanding of the standard	Conceptual model / Modelo conceptual
Level 1	Structuring / Estructuración	Complete documentation of the system	Defined architecture / Arquitectura definida
Level 2	Implementation / Implementación	Verifiable system with tests	Verifiable system / Sistema verificable
Level 3	Governance / Gobernanza	Auditable and controlled system in production	Certifiable system / Sistema certificable

K.2.2 Transition Principles / Principios de Transición

- Each level is cumulative. / Cada nivel es acumulativo.
- Cannot advance without fulfilling the previous level. / No se puede avanzar sin cumplir el nivel anterior.
- Each transition requires verifiable evidence. / Cada transición requiere evidencia verificable.

K.3 Implementation Framework (Modular) / Framework de Implementación (Modular)

Module / Módulo	Contents / Contenidos	Outcome / Resultado
A – Structural Compliance	Base Standard, required documents, technical glossaries	Documented and coherent architecture
B – Construction and Execution	Agent models, RAE, execution flows, tooling ecosystem	Implemented and operable system
C – Intelligence and Knowledge	Knowledge architecture, RAG strategies, structured information	System with structured cognitive capability
D – Evaluation and Observability	Metrics, benchmarking, observability, continuous validation	Measurable and evaluable system
E – Runtime Governance	Decision control, HITL, escalation rules, in-operation audit	Governed and auditable system in production

Modules are independent in adoption, but interdependent in full compliance. / Los módulos son independientes en adopción, pero interdependientes en conformidad completa.

K.4 Certification Framework / Framework de Certificación

K.4.1 Evaluation Dimensions / Dimensiones de Evaluación

Dimension / Dimensión	Evaluates / Evalúa
Structure / Estructura	Existence and quality of documents / Existencia y calidad de documentos
Verification / Verificación	Invariant coverage / Cobertura de invariantes
Traceability / Trazabilidad	State reconstructibility / Reconstruibilidad del estado
Governance / Gobernanza	Control and supervision of decisions / Control y supervisión de decisiones

K.4.2 Partial Certification / Certificación Parcial

Evaluation is permitted for: / La evaluación está permitida para:

- Complete system / Sistema completo
- Subsystems / Subsistemas
- Individual components / Componentes individuales

K.5 Governance Framework / Framework de Gobernanza

Component / Componente	Description / Descripción
Authority levels / Niveles de autoridad	Explicitly defined per agent / Definidos explícitamente por agente
Decision rules / Reglas de decisión	Documented, testable, traceable / Documentadas, testeables, trazables
Human intervention / Intervención humana	Conditions explicitly declared / Condiciones explícitamente declaradas
Escalation protocols / Protocolos de escalación	Defined for all unresolvable conflicts / Definidos para todos los conflictos irresolubles

No critical decision occurs without defined authority. / Ninguna decisión crítica ocurre sin autoridad definida.

K.6 Technology Compatibility / Compatibilidad Tecnológica

Conformity is structural, not technological. / La conformidad es estructural, no tecnológica.

The standard defines interfaces, expected behaviors, and invariants. It does NOT define specific tools, vendors, or languages.

El estándar define interfaces, comportamientos esperados e invariantes. NO define herramientas, proveedores o lenguajes específicos.

K.7 Packaging Framework / Framework de Empaquetado

Audience / Audiencia	Presentation / Presentación	Focus / Enfoque
Academic / Académico	Theoretical formalization, scientific publications	Conceptual models, formal proofs
Commercial / Comercial	Use cases, business value, organizational adoption	ROI, risk reduction, compliance
Operational / Operacional	Practical guides, step-by-step implementation	How-to, checklists, team playbooks

Content does not change, only its form of presentation. / El contenido no cambia, solo su forma de presentación.

ADDENDUM L — Universal Adoption & Derivation Framework

Addendum L – Universal Adoption & Derivation Framework

Purpose: Establish a practical framework that enables solution architects to translate the Canonical Design into all required standard artifacts systematically, independent of underlying technology and scalable from MVP to critical system.

Propósito: Establecer un framework práctico que habilite a arquitectos de soluciones traducir el Diseño Canónico en todos los artefactos requeridos sistemáticamente, independiente de tecnología subyacente.

The Canonical Design is the single source of truth. All other documents are derived from it through predictable transformations. Architecture is expressed in terms of FUNCTIONS (what is needed), not COMPONENTS (how it is implemented).

El Diseño Canónico es la única fuente de verdad. Todos los demás documentos se derivan de él mediante transformaciones predecibles. La arquitectura se expresa en términos de FUNCIONES (qué se necesita), no COMPONENTES (cómo se implementa).

L.1 Functional Equivalence Table / Tabla de Equivalencia Funcional

Standard Concept	Architectural Function (Agnostic)	Traditional Example	Cognitive/AI Example
Traceability (P1)	Immutable Recording Function	SQL audit table with mandatory INSERT on every state change	Event bus (Kafka, RabbitMQ) with cryptographically signed logs
Invariant Verification (P3)	Sufficiency Validation Function	Business validation before executing transaction	Context sufficiency validator (CSV) measuring semantic similarity
System State (Sec. 5)	Authoritative Reference Context	Transactional database (SSoT) reflecting real entity state	Knowledge corpus (vector store, graph) with verifiable evidence
Response Model (Sec. 6)	Structured Output Contract	JSON/XML with predefined fields; response always typed	Object with status: grounded/uncertainty, sources, trace
Governance (Sec. 7)	Automatic Escalation Function	Approval workflow if amount > threshold → manager signature	HITL rules when CSV confidence falls below threshold

Standard Concept	Architectural Function (Agnostic)	Traditional Example	Cognitive/AI Example
Compliance Metrics (CL-2/3)	Integrity Indicators	TCR = logged events / expected events URR = ungrounded responses / total GA = correct & grounded / domain queries FAR = false abstentions / total	URR, GA, PAR as defined in ECED thesis

L.2 Document Derivation Hierarchy / Jerarquía de Derivación de Documentos

Level	Document / Documento	Source / Fuente	Purpose / Propósito
0	Canonical Design	Single source of truth	Lead architect, invariant validation
1	Architecture Brief (Technical)	Projection of Canon onto operational context	Risk audits, regulatory compliance, technical presales
2	SSOT (Technology Stack)	Derivation from architectural functions to concrete tools	Development team, integrators, operations
3	C4 Model	Visualization of functional structure	Communication with technical and non-technical stakeholders
4	SDD	Detailed specification of interfaces, schemas, invariant tests	Implementers, testers
5	Executive Summary	Extraction of purpose, boundaries, compliance level	Board, clients, funding

Derivation Rule: Each document MUST include an explicit reference to the Canonical Design (version and date) from which it is derived. Any change to the Canon triggers revision of all derived documents.

Regla de Derivación: Cada documento DEBE incluir referencia explícita al Diseño Canónico (versión y fecha) del que se deriva. Cualquier cambio en el Canon dispara revisión de todos los documentos derivados.

L.3 Practical Application: Comparative Examples / Ejemplos Comparativos

Example A: Invoicing System (Traditional – Tier 2)

Canonical Design (excerpt):

- Purpose: Generate valid invoices from orders.
- **INV-F1:** No invoice can be issued without a confirmed order.
- **INV-F2:** Total amount must equal sum of items.
- Capability boundaries: No modification of issued invoices; only cancellation.

Document	Derivation / Derivación
Technical Brief	Validation flow (order → calculation → issuance) and access controls
SSOT	PostgreSQL, REST API with Node.js, logs in audit table
C4	Container diagram: database, invoicing service, message queue for notifications
SDD	Endpoint specs, JSON schemas, unit tests covering INV-F1 and INV-F2
Executive Summary	Low-risk system, CL-2 achievable with automated tests

Example B: Legal RAG Assistant (Cognitive – Tier 4)

Canonical Design (excerpt):

- Purpose: Answer legal queries based exclusively on an authorized corpus.
- **INV-2:** Every query produces complete traceability (search, fragments, response).
- **INV-3:** If corpus does not contain sufficient evidence, system must declare uncertainty (not invent).
- Capability boundaries: Cannot issue legal opinions without human supervision.

Document	Derivation / Derivación
Technical Brief	CSV specification (relevance thresholds, URR, PAR metrics) and escalation conditions to lawyer
SSOT	Qdrant (vector store), Llama 3 (LLM), Python, Kafka for events, Prometheus for metrics
C4	Containers: ingestion, retrieval, CSV, LLM, event bus, audit dashboard
SDD	API details, event structure, invariant tests (INV-2, INV-3) with synthetic data
Executive Summary	High-risk system, requires Global Sigma Tier 4 certification; CL-3 with active governance recommended

Example C: Industrial IoT Gateway (Embedded / Hybrid)

Canonical Design (excerpt):

- Purpose: Collect sensor telemetry and trigger actuation commands within safety bounds.
- **INV-G1:** No actuation command may be issued without a validated sensor reading within the last 500ms.
- **INV-G2:** All commands must be logged to non-volatile storage before execution.
- Capability boundaries: Cannot override hardware safety interlocks; cannot execute unvalidated firmware updates.

Document	Derivation / Derivación
Technical Brief	Sensor ingestion pipeline, validation window rules, actuation authorization flow
SSOT	C/C++ firmware, MQTT broker, SQLite on-device log, OTA update service with signing
C4	Context: cloud SCADA + edge gateway. Containers: sensor daemon, validation engine, command dispatcher, log store
SDD	Message schemas (MQTT topics), timing invariant tests, watchdog integration specs
Executive Summary	Safety-critical embedded system, CL-2 minimum, CL-3 required for regulated industries

L.4 Accelerating Development from MVP / Acelerando el Desarrollo desde MVP

1. MVP with Minimal Canon: Define purpose, basic functional invariants, and capability boundaries.
2. Automated (or semi-automated) Derivation: Technical Brief from invariants. SSOT by mapping functions to familiar tools. C4 built layer-by-layer. SDD structured with contracts defined in the Canon.
3. Continuous Verification: Every new invariant added to the Canon is automatically reflected in tests and audit criteria.

Quantitative benefit: In pilot projects, adopting this derivation reduced technical documentation time by 50–60% and eliminated inconsistencies between documents. / Beneficio cuantitativo: En proyectos piloto, adoptar esta derivación redujo el tiempo de documentación técnica un 50–60%.

L.5 Relation to Addendum K

Addendum K defines adoption levels (Level 0-3) and implementation modules. Addendum L complements K by providing the derivation mechanics.

Addendum	Establishes / Establece
K	WHEN and HOW to adopt the standard / CUÁNDO y CÓMO adoptar el estándar
L	WHICH artifacts to produce and HOW to derive them from the Canon / QUÉ artefactos producir y CÓMO derivarlos del Canon

ADDENDUM M — Minimum Document Templates / Plantillas Mínimas

Addendum M – Minimum Document Templates

Purpose: Provide skeleton templates for each required document (Section 3), enabling teams to immediately begin compliant documentation without starting from a blank page.

Propósito: Prover plantillas esqueleto para cada documento requerido (Sección 3), permitiendo que los equipos inicien documentación conforme de inmediato sin comenzar desde una página en blanco.

These templates are minimum starting points. Teams must populate them with system-specific content. Content does not change the requirements — only the templates change. / Estas plantillas son puntos de partida mínimos. Los equipos deben poblarlas con contenido específico del sistema.

M.1 Canonical Design Template

```
# CANONICAL DESIGN DOCUMENT
# System: [System Name]
# Version: [x.y] | Date: [YYYY-MM-DD] | Author: [Name]
# Canon-First: This document precedes all other system documents.

## 1. SYSTEM PURPOSE
[One clear sentence: What does this system exist to do?]
[Una oración clara: ¿Para qué existe este sistema?]

## 2. FUNDAMENTAL ENTITIES
| Entity | Description | Authority |
|-----|-----|-----|
| [Name] | [What it is] | [Who owns it] |

## 3. FUNCTIONAL INVARIANTS (INV-Fn)
| ID   | Invariant | Test | Status |
|-----|-----|-----|-----|
| INV-F1 | [System MUST always...] | [test-id] | [ ] |
| INV-F2 | [System MUST NEVER...] | [test-id] | [ ] |

## 4. VERIFICATION INVARIANTS (INV-Vn)
| ID   | Invariant | Mechanism |
|-----|-----|-----|
| INV-V1 | [Verifiable property...] | [How validated] |

## 5. BEHAVIOR RULES
- [Rule 1: If [condition] then [action]]
```

- [Rule 2: ...]

6. CAPABILITY BOUNDARIES (CB-n)

| CB-ID | The system CANNOT / MUST NOT | Reason |

|-----|-----|-----|

| CB-1 | [Prohibited action] | [Why] |

7. COMPLIANCE TARGET

Target CL: [] CL-0 [] CL-1 [] CL-2 [] CL-3

Justification: [...]

M.2 Architecture Brief Template

ARCHITECTURE BRIEF (TECHNICAL)

System: [System Name] | Derived from Canon v[x.y] dated [YYYY-MM-DD]

1. OPERATIONAL FLOW

[Describe the main operational sequence: trigger → process → output]

2. AGENT / MODULE ROLES

| Agent/Module | Role | Decisions it can make autonomously |

|-----|-----|-----|

| [Name] | [Role] | [Decisions] |

3. INTERACTION PATTERNS

[Synchronous / Asynchronous / Event-driven / Request-Response]

[Describe key interactions between agents/modules]

4. STATE TRANSITIONS

[State A] --[trigger]--> [State B]

[State B] --[trigger]--> [State C]

5. FUNCTIONAL CONFIGURATION

| Parameter | Default | Range | Notes |

|-----|-----|-----|

| [param] | [value] | [min-max] | [...] |

6. BOUNDARY MODEL

INPUTS accepted: [...]

OUTPUTS produced: [...]

EXTERNAL dependencies: [...]

PROHIBITED interactions: [...]

M.3 SSOT Template

SSOT — SINGLE SOURCE OF TRUTH

```
# System: [System Name] | Derived from Canon v[x.y] dated [YYYY-MM-DD]
```

1. TECHNOLOGY STACK

```
| Layer      | Technology | Version | Justification |
|-----|-----|-----|-----|
| Language   | [e.g. Python] | [3.12] | [Why] |
| Framework  | [e.g. FastAPI] | [0.x] | [Why] |
| Database    | [e.g. Postgres] | [16] | [Why] |
| Message Bus | [e.g. Kafka] | [3.x] | [Why] |
| Observability | [e.g. Prometheus] | [2.x] | [Why] |
```

2. ENVIRONMENTS

```
| Environment | Infrastructure | Configuration |
|-----|-----|-----|
| Development | [local/cloud] | [config notes] |
| Testing     | [...] | [...] |
| Production  | [...] | [...] |
```

3. CONFIGURATION CONSTRAINTS

- [Constraint 1: e.g., All secrets via environment variables, never hardcoded]
- [Constraint 2: ...]

M.4 C4 Model Checklist

C4 MODEL CHECKLIST

```
# System: [System Name] | Derived from Canon v[x.y]
```

LEVEL 1 — CONTEXT DIAGRAM []

Shows: System box + External users + External systems

Technology names: NOT required at this level

Validates: Canon §1 (Purpose) + §6 (Boundaries)

LEVEL 2 — CONTAINER DIAGRAM []

Shows: Internal containers (web app, API, DB, queue, etc.)

Technology names: REQUIRED (e.g., 'PostgreSQL 16', 'Node.js 20')

Validates: SSOT + Architecture Brief §3 (interaction patterns)

LEVEL 3 — COMPONENT DIAGRAM []

(Required only for critical containers)

Shows: Internal components of each critical container

Validates: SDD interfaces + Canon invariants

LEVEL 4 — CODE [] (Optional)

Shows: Class/module diagrams for critical components

Use only when architectural decision is at code level

M.5 SDD Template

```
# SOFTWARE DESIGN DOCUMENT (SDD)
# System: [System Name] | Derived from Canon v[x.y] dated [YYYY-MM-DD]

## 1. INTERFACES
### API Endpoint: [GET/POST/PUT/DELETE] /[path]
Request: { field: type, ... }
Response: { field: type, ... }
Invariants covered: [INV-F1, ...]

## 2. DATA SCHEMAS
### Event: [EventName]
{ id: uuid, timestamp: ISO8601, type: string, payload: {...} }

### Entity: [EntityName]
{ id: uuid, [fields], created_at, updated_at, version }

## 3. CONFIGURATION LOGIC
| Parameter | Type | Default | Validation Rule |
|-----|-----|-----|-----|
| [param] | [type] | [val] | [rule] |

## 4. VERIFICATION TESTS
| Test ID | Invariant | Input | Expected Output | Status |
|-----|-----|-----|-----|-----|
| T-001 | INV-F1 | [...] | [...] | [] |
| T-002 | INV-F2 | [...] | [...] | [] |
```

M.6 VCL Template

```
# VERIFICATION & COMPLIANCE LAYER (VCL)
# System: [System Name] | Compliance Target: CL-[n]

## 1. INVARIANT COVERAGE MATRIX
| Invariant | Test ID | Type | Status | Evidence |
|-----|-----|-----|-----|-----|
| INV-F1 | T-001 | Automated | [] | [path] |
| INV-F2 | T-002 | Manual | [] | [doc] |
| INV-V1 | T-003 | Automated | [] | [path] |

## 2. AUDIT MECHANISMS
| Mechanism | Implementation | Frequency | Owner |
|-----|-----|-----|-----|
| Event log | [tool/path] | Real-time | [team] |
| Audit report | [tool] | Weekly | [team] |

## 3. COMPLIANCE LEVEL ACHIEVED
```

- [] CL-0 — Documents exist at conceptual level
- [] CL-1 — All documents with sufficient detail
- [] CL-2 — Automated invariant tests + full traceability
- [] CL-3 — HITL + escalation + third-party audit

4. COVERAGE EVIDENCE

Test suite: [path/to/tests]

Last run: [YYYY-MM-DD]

Pass rate: [n/m] ([%])

Gaps identified: [list or 'None']

APPENDIX 1 — Equivalence with arc42

Appendix 1 – Equivalence with arc42

This appendix allows teams already using arc42 to find their entry path into the standard without abandoning their current framework. It does not replace any document of the standard and does not modify the Canon-First sequence.

Position of the Canonical Design relative to arc42

Framework	Approach
arc42	No document equivalent to the Canonical Design. System identity emerges from the 11 sections together — never explicitly declared before views and decisions.
This Standard	The Canonical Design PRECEDES everything. System identity is formally declared before any view, decision, or specification. arc42 becomes a projection of the Canonical.

The Canonical Design is the contribution that arc42 lacks. arc42 covers everything else — and the standard covers it as well.

Equivalence Table / Tabla de Equivalencia

arc42 Section	Standard Document / Documento del Estándar	Standard Section
01 – Introduction and goals	Canonical Design	§1.1 Definition + §2.1 Users
02 – Constraints	Canonical Design	INV-4 (Footprint) + CBs
03 – Context and scope	C4 Model	L1 Context Diagram
04 – Solution strategy	Architecture Brief	§2 OSL + Principles P1–P8
05 – Building block view (static)	C4 Model + SDD	L2/L3 + Contracts
06 – Runtime view (dynamic)	Architecture Brief	§2.5 Interaction flows
07 – Deployment view	SSOT	§4 Environment configuration
08 – Crosscutting concepts	Canonical Design	INV-2 (Traceability) + INV-6 + INV-8
09 – Architecture decisions	ADRs	Standard ADR format

arc42 Section	Standard Document / Documento del Estándar	Standard Section
10 – Quality requirements	Canonical Design	INV-V1 to INV-V6 + §9.2
11 – Risks and technical debt	Canonical Design	Capability Boundaries (Section 8) + gaps

What this Standard has that arc42 does not

Contribution / Contribución	Description / Descripción
Canonical Design	System identity formally declared before any view or decision. Verifiable invariants (INV-*). Non-negotiable Capability Boundaries (CB-*).
Compliance Levels CL-0 to CL-3	Progressive certification framework. Measurable conformity criteria. Path from CL-1 to CL-3 declared.
Verification Invariants (INV-V*)	Verifiability properties as part of design, not as post-QA.
State Authority Model	Who has authority over system state, how it is created, modified, and invalidated.
Canon-First Sequence	Canonical → Brief → SSOT → C4 → SDD. Code implements the Canonical. The Canonical is never derived from code.

Adoption Options for arc42 Teams

Option A – Full Adoption

1. Generate the Canonical Design first (the 5 questions + validation with the architect).
2. Project the Canonical onto arc42: each section of arc42 is derived from the Canonical.
3. Add what arc42 lacks: Compliance Levels, Verification Invariants, State Authority Model.
4. The team works with arc42. The Canonical is the source of truth.

Option B – Gradual Adoption

- Phase 1: Add the Canonical Design to the existing arc42 document set as 'Section 0 – System Identity'.
- Phase 2: Map the invariants of the Canonical to quality requirements (arc42 §10).
- Phase 3: Add Compliance Levels as maturity criteria to the existing process.
- Phase 4: Canon-First complete in the next project.

Option C – For Third-Party System Evaluation

5. arc42 §01 → extract system definition.
6. arc42 §02 → map to INV-4 and CBs.
7. arc42 §09 (ADRs) → validate against the standard.
8. arc42 §10 (quality) → map to INV-V*.
9. Determine CL level achieved (CL-0 if no explicit Canonical; CL-1 if arc42 complete; CL-2 if automated tests; CL-3 if third-party audit).

Note on Other Frameworks / Nota sobre Otros Frameworks

Framework	Relation to this Standard / Relación con este Estándar
arc42	Full equivalence — see table above
C4 Model	Adopted directly in the standard (L1–L4)
IEEE 1471 / ISO 42010	The Canonical satisfies the 'Architecture Description' requirements of the ISO standard
TOGAF	Compatible in architecture layers; the Canonical equates to Architecture Vision + Architecture Requirements
RUP / SAD	The Canonical + Brief cover the Software Architecture Document
Global Sigma	See Appendix 2 for full alignment mapping

APPENDIX 2 — Global Sigma Alignment

Appendix 2 – Alignment with Global Sigma Standard

Context: Global Sigma is an open certification framework for high-risk AI systems, authored by Dr. Paula Melissa Alfaro Montero (ORCID: 0009-0004-2849-5794), licensed under CC BY-SA 4.0. It operationalizes 12 fundamental principles (P1–P12) derived from international frameworks (EU AI Act, OECD, UNESCO, NIST, ISO/IEC) into verifiable, measurable requirements with reproducible audit methodology.

Repository: <http://github.com/lunalfaro/global-sigma-standard> | DOI: <https://doi.org/10.5281/zenodo.18866997>

Both frameworks share the same core value: verifiable, auditable, governable systems. They are complementary, not competing.

Ambos frameworks comparten el mismo valor central: sistemas verificables, auditables y gobernados. Son complementarios, no competidores.

Global Sigma: Key Innovation

Global Sigma distinguishes two tracks:

Track / Pista	Description / Descripción	Applicability
Technical Conformity	Voluntary; no immutability required; structural compliance	All AI systems; good practice
Certification	Mandatory for T3/T4 high-risk systems; requires immutability for forensic proof	High-risk AI per EU AI Act categories

Principle Alignment / Alineación de Principios

This Standard	Global Sigma Principle	Mapping / Mapeo
P1 – Traceability	P2 – Transparency; P7 – Accountability	Full alignment: immutable recording function maps directly to Sigma audit trail requirements
P2 – State Consistency	P3 – Reliability & Robustness	Alignment: Sigma's reliability requirements directly depend on consistent state representation
P3 – Verifiability	P4 – Explainability; P8 – Human Oversight	Alignment: Sigma's explainability and HITL requirements map to invariant-based verification

This Standard	Global Sigma Principle	Mapping / Mapeo
P4 – Governance	P8 – Human Oversight; P10 – Accountability	Full alignment: Sigma's HITL and escalation patterns are a superset of P4
P5 – Bounded Capability	P5 – Privacy; P6 – Safety; P11 – Fairness	Alignment: Sigma's prohibited use cases and safety constraints are expressed as CB-* in this standard

Compliance Level ↔ Global Sigma Tier Mapping

This Standard	Global Sigma Tier	Notes / Notas
CL-0 / CL-1	T1 – Minimal Risk	Documented systems with basic structural compliance
CL-1 / CL-2	T2 – Limited Risk	Documented + partially verifiable systems
CL-2	T3 – High Risk	Automated tests + full traceability; Sigma requires additional immutability
CL-3	T4 – Critical Risk	Full governance + HITL + third-party reproducible audit; maps to Sigma mandatory certification track

Document Mapping / Mapeo de Documentos

This Standard Document	Global Sigma Equivalent / Equivalente	Notes
Canonical Design	System Identification Form + Risk Classification	Canon's invariants map to Sigma's risk tier determination criteria
Architecture Brief	Technical Risk Assessment	Sigma requires similar operational flow documentation for T3/T4
SSOT	Conformity Assessment Technical File	Sigma's EU AI Act alignment requires explicit technology documentation
C4 Model	System Architecture Diagram (Annex D)	Sigma recommends C4 or equivalent for T2+
SDD	Technical Documentation (Art. 11, EU AI Act)	Direct mapping for regulated AI systems
VCL	Audit Evidence Package + Conformity Declaration	Sigma's reproducible audit requirement maps directly to VCL

Combined Use / Uso Combinado

For teams building high-risk AI systems, the recommended approach is to use both frameworks in combination:

- 10. **Start with this Standard:** Define Canonical Design, derive all required documents, achieve CL-2.
- 11. **Map to Global Sigma Tier:** Classify system risk tier using Global Sigma T1–T4 methodology.
- 12. **Apply Sigma-specific requirements:** For T3/T4 systems, add immutability, HITL, and forensic evidence layers per Sigma.
- 13. **Unified audit:** VCL serves as base evidence package; extend with Sigma's Conformity Declaration for regulatory submission.

Para equipos construyendo sistemas de IA de alto riesgo, se recomienda usar ambos frameworks en combinación: este estándar provee la base arquitectónica; Global Sigma provee el framework de certificación regulatoria.

Governance Alignment

Both frameworks share a strong commitment to governance independence:

Principle	This Standard	Global Sigma
Author independence	Standard is technologically agnostic; no vendor lock-in	Governance transferred to Guardian Foundation; author decoupled from governance
Open access	No licensing restrictions on the standard itself	CC BY-SA 4.0; public asset
Audit reproducibility	CL-3 requires third-party reproducible audit	Mandatory for T3/T4 certification
Non-capture	No prescribed tools or vendors	Rule-Maker (Foundation) and Tool-Provider (author's firm) kept independent

CONSOLIDATED GLOSSARY / GLOSARIO CONSOLIDADO

Consolidated Glossary / Glosario Consolidado

This glossary defines all key terms used across the standard and its addenda. Terms are listed alphabetically with both English and Spanish definitions.

Este glosario define todos los términos clave usados en el estándar y sus addenda. Los términos se listan alfabéticamente con definiciones en inglés y español.

Term (EN/ES)	Definition / Definición
Agent / Agente	An autonomous or semi-autonomous module that can make decisions and take actions within defined authority boundaries. / Módulo autónomo o semi-autónomo que puede tomar decisiones y acciones dentro de límites de autoridad definidos.
Architecture Brief / Brief Arquitectónico	The document that instantiates the Canonical Design in a concrete functional context. It defines operational flow, roles, state transitions, and boundary model. / Documento que instancia el Diseño Canónico en un contexto funcional concreto.
Audit / Auditoría	The process of verifying that a system complies with the requirements of this standard, producing reproducible evidence. / Proceso de verificación de que un sistema cumple con los requisitos de este estándar, produciendo evidencia reproducible.
C4 Model / Modelo C4	A hierarchical architecture visualization framework with four levels: Context (L1), Container (L2), Component (L3), Code (L4). / Framework de visualización arquitectónica jerárquica con cuatro niveles.
Canonical Design / Diseño Canónico	The primary document that defines the essence of a system: its purpose, entities, invariants, rules, and boundaries. The single source of truth that precedes all other documents. / El documento primario que define la esencia del sistema. La única fuente de verdad que precede a todos los demás documentos.
Capability Boundary (CB) / Límite de Capacidad	An explicit definition of what a system cannot or must not do. Part of the system's auditable identity. / Definición explícita de lo que un sistema no puede o no debe hacer. Parte de la identidad auditable del sistema.
Canon-First / Canon-Primero	The design sequence in which the Canonical Design is created before any other document, and all other documents are derived from it. / La secuencia de diseño en la que el Diseño Canónico se crea antes que cualquier otro documento.
Compliance Level (CL) / Nivel de Conformidad	A classification (CL-0 to CL-3) that indicates the degree to which a system satisfies the requirements of this standard. / Clasificación (CL-0 a CL-3) que indica el grado en que un sistema satisface los requisitos de este estándar.
CSV (Context Sufficiency Validator) / CSV (Validador de Suficiencia de Contexto)	A component that evaluates whether retrieved context contains sufficient evidence to generate a response before invoking an LLM. Operates as a governance gatekeeper. / Componente que evalúa si el contexto recuperado contiene evidencia suficiente para generar

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	una respuesta antes de invocar un LLM. Actúa como guardián de la gobernanza.
Event / Evento	An immutable record of something that happened in the system, used as a primary traceability artifact. / Registro inmutable de algo que ocurrió en el sistema, usado como artefacto primario de trazabilidad.
Governance / Gobernanza	The explicit definition of decision authority within a system: which agents can make which decisions, and when human intervention is required. / Definición explícita de autoridad de decisión dentro de un sistema.
HITL (Human-in-the-Loop)	A governance pattern where human supervision is mandatory for specific decisions or conditions. Required for CL-3. / Patrón de gobernanza donde la supervisión humana es mandatoria para decisiones o condiciones específicas. Requerido para CL-3.
Invariant / Invariante	A property that must always be true for a system to be considered correct. Has a unique identifier and at least one validation mechanism. / Propiedad que debe ser siempre verdadera para que un sistema sea considerado correcto.
Modular Adoption / Adopción Modular	The practice of implementing the standard incrementally through defined modules (A–E), enabling progressive compliance without sacrificing integrity. / Práctica de implementar el estándar de forma incremental mediante módulos definidos (A–E).
MVP (Minimum Viable Product)	The simplest system implementation that satisfies the minimum Canonical Design requirements and achieves at least CL-0. / La implementación de sistema más simple que satisface los requisitos mínimos del Diseño Canónico y alcanza al menos CL-0.
RAG (Retrieval-Augmented Generation) / RAG (Generación Aumentada por Recuperación)	An architecture that grounds LLM generation in retrieved documents. The standard requires that such grounding be verifiable and that the system declare uncertainty when evidence is insufficient. / Arquitectura que fundamenta la generación del LLM en documentos recuperados. El estándar requiere que dicha fundamentación sea verificable y que el sistema declare incertidumbre cuando la evidencia es insuficiente.
RFI/RFP	Request for Information / Request for Proposals — the formal package required for system audit submission. / Paquete formal requerido para la presentación de auditoría del sistema.
SDD (Software Design Document)	The document providing detailed implementation specifications: interfaces, schemas, APIs, and invariant tests. / Documento que provee especificaciones de implementación detalladas.
SSOT (Single Source of Truth)	The document that records concrete technological decisions: stack, versions, environments, and justifications. / Documento que registra decisiones tecnológicas concretas.
State / Estado	The complete set of information that represents the current condition of a system at a given moment. Must be consistent and reconstructible. / Conjunto completo de información que representa la condición actual de un sistema en un momento dado.

Term (EN/ES)	Definition / Definición
Traceability / Trazabilidad	The ability to explain any system behavior by reference to observable artifacts (events, logs, state). / Capacidad de explicar cualquier comportamiento del sistema por referencia a artefactos observables.
VCL (Verification & Compliance Layer)	The document providing evidence that the system is correct and auditable: invariant tests, audit mechanisms, compliance level, coverage evidence. / Documento que provee evidencia de que el sistema es correcto y auditable.
Verifiability / Verificabilidad	The property of a system whose behavior can be tested against defined invariants, each with at least one documented validation mechanism. / Propiedad de un sistema cuyo comportamiento puede testearse contra invariantes definidos.

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Architectural Standard for Software Systems v1.2
BASE STANDARD — FROZEN · Bilingüe ES/EN · Professional Reference
Complemented by Canonical Design Guide and Addenda A–M